Detecting Forward-Looking Market Alpha (α) vs. Priced-In Consensus Echo: Multi-Hop Supply-Chain, Footnote, and Procurement Graph Verification

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Abstract & Summary: Why unassisted LLMs and homogeneous agent swarms spend 76%–82% of their output repeating public headlines already priced into the market, and how competing-hypothesis scenario trees fix it.

1. THE PRICED-IN CONSENSUS TRAP IN FINANCIAL LLM RESEARCH

When an institutional investor or research analyst asks an LLM to evaluate a company or macro catalyst, the most common failure is not hallucination—it is **consensus echo**. Public earnings headlines, sell-side summaries, and management guidance are the highest-frequency tokens in pre-training and web-search retrieval. Yet in any liquid market, facts that every participant repeats in the first paragraph of a summary are already priced into the asset.

Worse, when practitioners deploy a homogeneous 5-agent LLM swarm without independent priors or structured disconfirmation, the swarm exhibits **echo amplification**: the share of tokens spent repeating priced-in consensus rises from **76.0% (single LLM)** to **82.0% (5-agent swarm)**, while second- and third-order supply-chain risks buried in footnotes and subcontractor registries are smoothed away during summarization.

2. THE THREE-HOP STRUCTURAL ALPHA BENCHMARK (24 HISTORICAL & SYNTHETIC CASES)

We constructed a 24-case benchmark (16 true structural-divergence cases paired with 8 matched benign working-capital twins) where the true forward-looking signal requires connecting three dispersed hops:

- **Hop 1 (Public Surface Filing):** Top-line revenue beat (+18% YoY) and reiterated full-year guidance that anchors consensus ratings at *Outperform*.
- **Hop 2 (Footnote / Working-Capital Divergence):** Days Sales Outstanding (DSO) jumping from 41 to 79 days alongside a 64% drop in raw resin inventory at a single-source subcontractor following a municipal water-permit restriction.
- **Hop 3 (Downstream Customer / Procurement Signal):** Primary buyers quietly qualifying a next-generation substitute (e.g., glass-core substrates) for Q2 delivery.

Evaluation Arm (24 Multi-Hop Cases)	Priced-In Consensus Echo Share	2nd/3rd-Order Catalyst Recall	Matched Benign Twin Precision	Scenario Calibration (`1 - Brier`)
Arm 1: Unassisted Single Frontier LLM	**76.0%**	**25.0% (4/16)**	62.5%	**0.31**
Arm 2: Homogeneous 5-Agent Research Swarm	**82.0% (Echo Collapse)**	**31.3% (5/16)**	62.5%	**0.29**
Arm 3: Naive “Short-Seller / Bearish” Prompt	54.0%	68.8% (11/16)	**25.0% (75% False Alarms)**	0.44
Arm 4: Baltor Multi-Hop Graph + Competing-Hypothesis Tree	**14.0%**	**93.8% (15/16)**	**87.5% (7/8 Cleared)**	**0.88**

Why the Same Architecture Solves Both Counter-Trafficking and Market Alpha: In both domains, the surface narrative is engineered to look reassuringly routine (a “zero-fee” employment contract or an “+18% YoY revenue beat”), while the decisive reality lives in the multi-hop constraint graph (the 68% APR debt novation in Hong Kong, or the 79-day DSO spike + subcontractor water-permit halt). Stripping consensus boilerplate and forcing explicit Protagonist vs. Antagonist scenario trees solves both problems with the same verification engine.